

Textbook monopoly model

A monopolist (single seller) produces a single good. The cost function is $c : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is C^1 with $c' > 0$ and satisfies $c(0) = 0$.

The market demand function is $d : \mathbb{R}_+ \rightarrow \mathbb{R}_+$. (If $\lim_{p \rightarrow 0} d(p) = \infty$, then restrict its domain to be $\mathbb{R}_{++}$.) You should think of this demand as generated by consumers, each of whom has preferences that are quasilinear in the *same* good, one which can be any good *other* than the one the monopolist produces. It follows that demand does not depend on income. The demand d is *continuous* and *nonincreasing* on $\mathbb{R}_{++}$; and on $\{p | d(p) > 0\}$ —the set of prices for which demand is positive—it is C^1 with $d' < 0$.

As an example, the demand $d(p) = \max\{a - bp, 0\}$ for $a, b > 0$ satisfies all these conditions. Demand is 0 for $p \geq a/b$. (I suggest you draw a picture of it.)

The monopolist's problem is

$$\pi = \max_{(q,p) \geq 0} pq - c(q) \quad (1)$$

$$\text{subject to } q \leq d(p) \quad (2)$$

In words: the monopolist chooses price and quantity to maximize profit, subject to the participation constraint (or “no-slavery condition”) that the quantity cannot be larger than what consumers demand at the price. The constraint (2) must bind at any solution with $q > 0$ (why?), so we can (as a simple application of the Principle of Optimality) write the problem as

$$\pi = \max_{p \geq 0} pd(p) - c(d(p)). \quad (3)$$

This formulation is usually best when the emphasis is on price (as in #3 on ASMT VII).

For $q \geq 0$ define $\tilde{p}(q) = d^{-1}(q)$, the *inverse demand function*. (If $\lim_{q \rightarrow 0} \tilde{p}(q) = \infty$, then restrict its domain to be $\mathbb{R}_{++}$; otherwise define $\tilde{p}(0)$ to be the smallest price such that $d(p) = 0$: $\tilde{p}(0) = \lim_{q \rightarrow 0} d^{-1}(q)$.) As an example, for the demand $d(p) = \max\{a - bp, 0\}$, the inverse demand is

$$\tilde{p}(q) = \max \left\{ 0, \frac{a}{b} - \frac{q}{b} \right\}.$$

Another way to write the firm's problem is

$$\pi = \max_{q \geq 0} \tilde{p}(q)q - c(q). \quad (4)$$

This formulation is usually best when the emphasis is on quantity (as in #4 on ASMT VII).

We will compare the choice of q for the firm using formulation (4) with a benevolent social planner who chooses $q \geq 0$ to maximize social surplus, $\int_0^q \tilde{p}(r)dr - c(q)$.